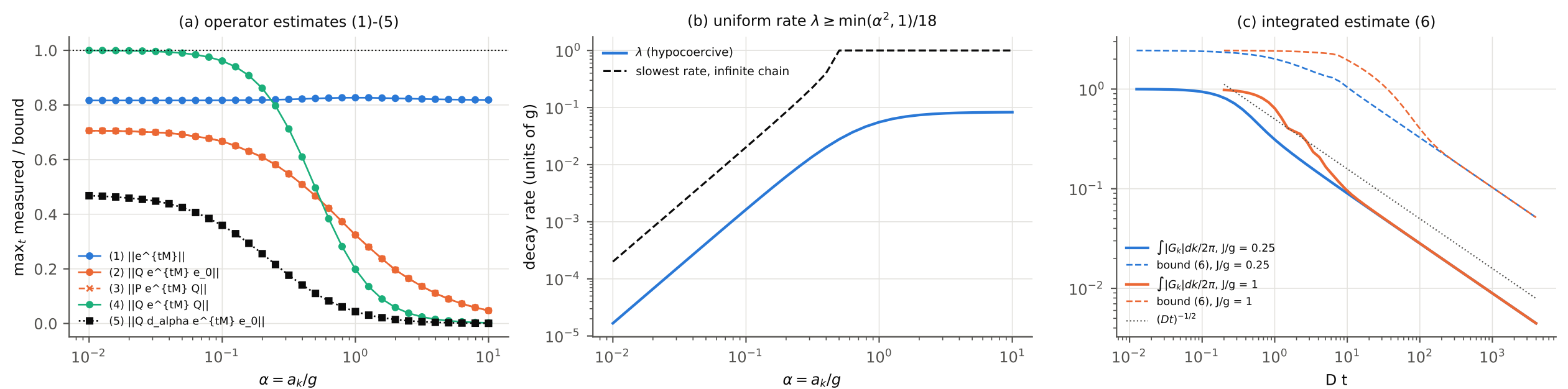
(a) operator estimates (1)-(5)
$\max_t$ measured / bound
$\alpha = a_k/g$
(1) ||e^{tM}||
(2) ||Q e^{tM} e_0||
(3) ||P e^{tM} Q||
(4) ||Q e^{tM} Q||
(5) ||Q d_alpha e^{tM} e_0||
(b) uniform rate $\lambda \geq \min(\alpha^2, 1)/18$
decay rate (units of g)
λ (hypocoercive)
slowest rate, infinite chain
$\alpha = a_k/g$
(c) integrated estimate (6)
$\int |G_k| dk/2\pi$, J/g = 0.25
bound (6), J/g = 0.25
$\int |G_k| dk/2\pi$, J/g = 1
bound (6), J/g = 1
$(Dt)^{-1/2}$
D t